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Zain Dana Harper

Curriculum Vitae | Cross-domain Systems Engineer

Seattle, Washington | Open to remote, hybrid, onsite, contract, part-time, and project-based work
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Profile

Seattle-based systems engineer whose public work spans compilers and language tooling, real-time graphics and color science, AI-agent infrastructure, verification and research tooling, technical documentation, and operations. Public career materials describe an independent engineering practice since 2023, grounded in earlier technical support, field operations, and freelance technical writing work.

The throughline is practical systems work: enter ambiguity, map the moving parts, build the working surface, preserve evidence, and leave an artifact another person can inspect or use. Overall programming experience includes roughly thirteen years of hobby, self-directed, and exploratory development; the serious public and release-driven engineering arc begins in 2023.

Engineering domains

AI-agent infrastructure and evaluation. Multi-provider and local-endpoint routing, tool-use loops, MCP tools, task decomposition, multi-agent orchestration, worker/verifier separation, provenance, replayable ledgers, evidence intake, re-checkable receipts, and public proof surfaces.

Compilers and language tooling. Lexing, parsing, AST and intermediate representations, type checking, code generation, typed effects, ownership and lifetime analysis, native C output, shader output, editor tooling, and public package surfaces.

Graphics, color, and native systems. D3D11/HLSL rendering, proxy-DLL architecture, post-processing, shared-memory IPC, GPU trace validation, color spaces and appearance models, HDR and tone mapping, ICC/VCGT and 3D-LUT workflows, perceptual color difference, DDC/CI, and display characterization.

Documentation, compliance, and operations. API and developer documentation, implementation guides, support material, proposals, release notes, operational playbooks, structured authoring, customer communication, site assessment, scheduling, safety judgment, and vendor coordination.

Selected public systems

Project Telos | **public ecosystem** | **early R&D and versioned tools**

A public line of tools for model-era engineering: shared state, repository intelligence, evidence intake, multi-agent orchestration, evaluation, verification, release hygiene, documentation, and interactive demonstrations. The portfolio intentionally distinguishes released packages, public source packages, release candidates, experimental branches, and legacy surfaces.

Index | index-graph 2.9.0 | beta

Repository inventory, dependency graph, documentation atlas, context packs, snapshots, drift checks, and route hints for large multi-repository workspaces.

Gather | gather-engine 1.6.1 | tagged public release

Research intake for web, feeds, documents, PDFs, browser and OCR paths, and authenticated APIs, with provenance records describing how each item was obtained.

Forum | forum-engine 1.13.0 | versioned public package

Model-agnostic orchestration for agents. Records routing, plans, task results, verdicts, and checkpoints in a replayable ledger.

Crucible | crucible-bench 1.2.0 | versioned public package

Evaluation workflows that separate the worker from a verifier reviewing the original criterion and final artifact. Supports thesis evaluation, weak-axis refinement, and preserved outcomes.

EMET | 1.0.0 on default public main | frozen specification

Byte-level witnessing that re-derives a file from source bytes and reports MATCH, DRIFT, or UNVERIFIABLE. Later source work exists on a non-default branch and is not represented as the current release.

BuildLang | 1.2.0 default-main source version

Rust compiler and systems-language project with typed effects, ownership and lifetime analysis, a primary C backend, shader outputs, editor support, and a public crates.io surface. Additional backends and self-hosting remain explicitly experimental.

Relay, Plexus, and Mneme | 0.1.0 public source packages

Related zero-runtime-dependency packages for model-endpoint coding-agent routing, toolchain auto-wiring, and provenance-carrying agent memory. Public source availability is not represented as registry publication where a registry release has not been verified.

Studio Engine | public browser-native creative system

Interactive and generated visual surfaces used to present systems work as inspectable artifacts. Includes Canvas/WebGL work, deterministic capture paths, design-system assets, and portfolio media generation.

Real-time graphics work

Public HLSL/C++ graphics work includes a Skyrim post-processing project whose current public career materials report more than 900,000 downloads, plus D3D11/HLSL systems involving tone mapping, TAA, SSR, SSGI, GTAO, volumetrics, GPU traces, and read-only shared-memory bridges.

Upstream open-source contributions

Merged into public repositories maintained by others | 22 pull requests across 19 repositories

Defect repair, test coverage, and documentation accepted upstream in projects including Datasette, tomlkit, pydantic-ai, DeepEval, pydash, and grimp. Representative merged work: HTTP status handling for malformed composite-key row URLs, a TOML round-trip bug where an array of tables swallowed the next sibling key, a

metric crash on unhashable tool outputs, missing module and layer validation errors, and lint failures blocking a red main branch. A second group adds test and conformance coverage to agent and orchestration projects: seeded determinism runs, mid-task recovery fixtures, edge-case fixtures, and evaluation benchmark suites.

Open and awaiting maintainer review | 14 pull requests across 13 repositories

Open pull requests in projects including the Model Context Protocol Inspector and Python SDK, Poetry, Drizzle ORM, and LLM. These are open, not accepted, and are counted separately from the merged total.

Professional experience

Independent Systems Engineer / Founder-Builder | Project Telos and HarperZ9 Seattle / remote | 2023-present

- Build and maintain the public systems portfolio across Python, Rust, C++, HLSL, TypeScript/JavaScript, and web-native surfaces.
- Own architecture, implementation, test strategy, documentation, packaging, integration, release notes, demos, public claims, and maintenance planning.
- Use coordinated coding agents for parallel discovery, implementation, and review while keeping first-party architecture, integration, verification, and release decisions explicit.
- Design capability-environment improvements alongside product work: reusable benchmarks, adapters, release checks, proof indexes, recording paths, and documentation that make the next cycle faster and more reliable.

Freelance Technical Writer / GRC Documentation / Product Operations Remote | 2017-present

- Produce API and implementation guides, security and compliance documentation, proposals, RFP and grant material, release notes, support content, and operational playbooks.
- Work with NIST 800-171, CMMC readiness, SOC 2, ISO 27001, DFARS, incident response, and audit-support concepts in a technical-writing capacity.
- Translate implementation detail into material usable by engineers, operators, customers, executives, and nontechnical stakeholders.
- Keep client names, private deliverables, compliance evidence, and contract terms outside the public portfolio.

Operations and Commercial Arboriculture Lead | Family business Seattle area | 2015-present

- Lead client intake, estimates, site assessment, scheduling, safety judgment, crew and vendor coordination, proposals, and customer communication.
- Coordinate work under changing physical conditions with practical risk assessment, clear handoffs, and accountable completion.
- Carry field-tested operating discipline into software and research systems.

Technical Networking Support, Xbox Division | Microsoft Redmond, Washington | 2013-2014

- Diagnosed TCP/IP, DNS, NAT, firewall, router, and account-adjacent console networking issues across phone and chat support.
- Documented repeatable resolutions and translated technical fixes into clear customer guidance.

Technical skills

Languages: Python, Rust, C++, TypeScript/JavaScript, Lua, HLSL, C#, PowerShell, Bash.

Development and delivery: Git, GitHub, pytest, Node test tooling, GitHub Actions, Linux, CMake, vcpkg, package metadata, release notes, CLI design, MCP/tool surfaces, static sites, Canvas/WebGL.

Systems: compiler pipelines, type systems, effects, ownership/lifetimes, repository graphs, state machines, ledgers, provenance, shared-memory IPC, graphics injection, GPU workflows, evidence contracts.

Documentation: developer and API documentation, implementation and support guides, proposals, release documentation, technical research synthesis, troubleshooting, compliance and audit-support writing.

Work preferences

Remote preferred; open to onsite or hybrid work, contract, full-time, part-time, and project-based engagements, with flexibility for travel or field-facing work where appropriate. Strong fits include compiler and language tooling, graphics and color, AI/ML infrastructure, research operations, security and verification, technical writing, implementation, QA, product operations, and domain-agnostic systems work.

Public links

Main site: <https://harperz9.github.io> Portfolio: <https://harperz9.github.io/portfolio.html> Research: <https://harperz9.github.io/research.html> Repositories: <https://github.com/HarperZ9> ORCID: <https://orcid.org/0009-0001-7175-5393>

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